

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

Product name : Ethylene

Product code : X2111, X2112, X2270, X2273, Q9248, E7000

CAS-No. : 74-85-1

1.2 Identified relevant uses of the substance or mixture and restrictions on use

Recommended use of the chemical and restrictions on use

Recommended use : Base chemical., Raw material for use in the chemical industry.

Restrictions on use : This product must not be used in applications other than the above without first seeking the advice of the supplier.

1.3 Details of the supplier of the safety data sheet

Manufacturer or supplier's details

Manufacturer/Supplier : **Shell Chemicals Europe B.V.**
PO Box 2334
3000 CH Rotterdam
Netherlands

Telephone : +31 (0)10 231 7000

Telefax : +31 (0)10 231 7180

1.4 Emergency telephone number

Emergency telephone number : +31 (0)10 231 7393
UAT for SPS2020 - New ER number

2. HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Flammable gases : Category 1A
H220: Extremely flammable gas.

Gases under pressure : Compressed gas
H280: Contains gas under pressure; may explode if heated.

Specific target organ toxicity - single exposure : Category 3 (Narcotic effects)
H336: May cause drowsiness or dizziness.

2.2 Label elements

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Hazard pictograms	:	
Signal word	:	Danger
Hazard statements	:	PHYSICAL HAZARDS: H220 Extremely flammable gas. H280 Contains gas under pressure; may explode if heated. HEALTH HAZARDS: H336 May cause drowsiness or dizziness. ENVIRONMENTAL HAZARDS: Not classified as environmental hazard according to CLP criteria.
Precautionary statements	:	Prevention: P210 Keep away from heat/ sparks/ open flames/ hot surfaces. No smoking. P243 Take precautionary measures against static discharge. P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray. Response: P377 Leaking gas fire: Do not extinguish, unless leak can be stopped safely. P381 Eliminate all ignition sources if safe to do so. P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Storage: P410 + P403 Protect from sunlight. Store in a well-ventilated place. Disposal: P501 Dispose of contents/ container to an approved waste disposal plant.

2.3 Other hazards

May form flammable/explosive vapour-air mixture. This material is a static accumulator. Even with proper grounding and bonding, this material can still accumulate an electrostatic charge. If sufficient charge is allowed to accumulate, electrostatic discharge and ignition of flammable air-vapour mixtures can occur. This material is shipped under pressure. High gas concentrations will displace available air; unconsciousness and death may occur suddenly from lack of oxygen. Rapid release of gases which are liquids under pressure may cause frost burns of exposed tissues (skin, eye) due to evaporative cooling.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Substance

3.1 Substances

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Hazardous components

Chemical name	CAS-No. EC-No. Registration number	Classification (REGULATION (EC) No 1272/2008)	Concentration (% w/w)
ethylene	74-85-1	Flam. Gas 1A; H220 Press. Gas Compr. Gas; H280 STOT SE 3; H336	>= 99,9

For explanation of abbreviations see section 16.

4. FIRST-AID MEASURES

4.1 Description of first aid measures

- General advice : Not expected to be a health hazard when used under normal conditions.
- If inhaled : Remove to fresh air. If rapid recovery does not occur, transport to nearest medical facility for additional treatment.
- In case of skin contact : Slowly warm the exposed area by rinsing with warm water. Transport to the nearest medical facility for additional treatment.
- In case of eye contact : Slowly warm the exposed area by rinsing with warm water. Transport to the nearest medical facility for additional treatment.
- If swallowed : In general no treatment is necessary unless large quantities are swallowed, however, get medical advice.

4.2 Protection of first-aiders

- Protection of first-aiders : When administering first aid, ensure that you are wearing the appropriate personal protective equipment according to the incident, injury and surroundings.

4.3 Most important symptoms and effects, both acute and delayed

- Most important symptoms and effects, both acute and delayed : Breathing of high vapour concentrations may cause central nervous system (CNS) depression resulting in dizziness, light-headedness, headache, nausea and loss of coordination. Continued inhalation may result in unconsciousness and death.
Rapid release of gases which are liquids under pressure may cause frost burns of exposed tissues (skin, eye) due to evaporative cooling.
No specific hazards under normal use conditions.

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Notes to physician	: Ingestion may result in nausea, vomiting and/or diarrhea. : IMMEDIATE TREATMENT IS EXTREMELY IMPORTANT! Call a doctor or poison control center for guidance. Treat symptomatically. Potential for cardiac sensitisation, particularly in abuse situations. Hypoxia or negative inotropes may enhance these effects. Consider: oxygen therapy.
--------------------	---

5. FIRE-FIGHTING MEASURES

5.1 Extinguishing media

Suitable extinguishing media	: Shut off supply. If not possible and no risk to surroundings, let the fire burn itself out.
Unsuitable extinguishing media	: Data not available

5.2 Special hazards arising from the substance or mixture

Specific hazards during firefighting	: Sustained fire attack on vessels may result in a Boiling Liquid Expanding Vapor Explosion (BLEVE). Contents are under pressure and can explode when exposed to heat or flames. As the vapours become lighter than air, the vapours may reach ignition sources at ground or elevated locations.
--------------------------------------	--

5.3 Recommendations for fire-fighters

Specific extinguishing methods	: Standard procedure for chemical fires. Clear fire area of all non-emergency personnel. Keep adjacent containers cool by spraying with water.
Special protective equipment for firefighters	: Proper protective equipment including chemical resistant gloves are to be worn; chemical resistant suit is indicated if large contact with spilled product is expected. Self-Contained Breathing Apparatus must be worn when approaching a fire in a confined space. Select fire fighter's clothing approved to relevant Standards (e.g. Europe: EN469).

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions	: Observe the relevant local and international regulations Notify authorities if any exposure to the general public or the environment occurs or is likely to occur. Local authorities should be advised if significant spillages cannot be contained. Avoid contact with skin, eyes and clothing. Isolate hazard area and deny entry to unnecessary or
----------------------	---

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

unprotected personnel.
Do not breathe fumes, vapour.
Do not operate electrical equipment.

6.2 Environmental precautions

Environmental precautions : Shut off leaks, if possible without personal risks. Remove all possible sources of ignition in the surrounding area and evacuate all personnel. Attempt to disperse the gas or to direct its flow to a safe location for example by using fog sprays. Take precautionary measures against static discharge. Ensure electrical continuity by bonding and grounding (earthing) all equipment. Monitor area with combustible gas meter.

6.3 Methods and material for containment and cleaning up

Methods and materials for containment and cleaning up : Allow to evaporate.
Attempt to disperse the vapour or to direct its flow to a safe location, for example by using fog sprays. Otherwise treat as for small spillage.

6.4 Reference to other sections

For guidance on selection of personal protective equipment see Section 8 of this Safety Data Sheet., Vapour may form an explosive mixture with air., For guidance on disposal of spilled material see Section 13 of this Safety Data Sheet.

7. HANDLING AND STORAGE

7.1 Precautions for safe handling

General Precautions : Avoid breathing of or direct contact with material. Only use in well ventilated areas. Wash thoroughly after handling. For guidance on selection of personal protective equipment see Section 8 of this Safety Data Sheet.
Use the information in this data sheet as input to a risk assessment of local circumstances to help determine appropriate controls for safe handling, storage and disposal of this material.

Advice on safe handling : This product is intended for use in closed systems only.
Extinguish any naked flames. Do not smoke. Remove ignition sources. Avoid sparks.
Avoid inhaling vapour and/or mists.
Avoid contact with skin, eyes and clothing.
Use local exhaust ventilation if there is risk of inhalation of vapours, mists or aerosols.
Bulk storage tanks should be diked (bunded).

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Properly dispose of any contaminated rags or cleaning materials in order to prevent fires.
Even with proper grounding and bonding, this material can still accumulate an electrostatic charge.
If sufficient charge is allowed to accumulate, electrostatic discharge and ignition of flammable air-vapour mixtures can occur.
Be aware of handling operations that may give rise to additional hazards that result from the accumulation of static charges.
These include but are not limited to pumping (especially turbulent flow), mixing, filtering, splash filling, cleaning and filling of tanks and containers, sampling, switch loading, gauging, vacuum truck operations, and mechanical movements.
These activities may lead to static discharge e.g. spark formation.
Restrict line velocity during pumping in order to avoid generation of electrostatic discharge (≤ 1 m/s until fill pipe submerged to twice its diameter, then ≤ 7 m/s). Avoid splash filling.
Do NOT use compressed air for filling, discharging, or handling operations.

Avoidance of contact : Strong oxidising agents.
Hydrochloric acid, hydrogen bromide and nitrogen oxides.

Product Transfer : Refer to guidance under Handling section.

7.2 Conditions for safe storage, including any incompatibilities

Other data : Vapours from tanks should not be released to atmosphere.
Breathing losses during storage should be controlled by a suitable vapour treatment system.
Electrostatic charges will be generated during pumping.
Electrostatic discharge may cause fire. Ensure electrical continuity by bonding and grounding (earthing) all equipment to reduce the risk.
The vapours in the head space of the storage vessel may lie in the flammable/explosive range and hence may be flammable.
Must be stored in a diked (bunded) well-ventilated area, away from sunlight, ignition sources and other sources of heat.
Keep away from aerosols, flammables, oxidizing agents, corrosives and from products harmful or toxic to man or to the environment.

Packaging material : Suitable material: For containers, or container linings use mild steel, stainless steel.

7.3 Specific end use(s)

Specific use(s) : Base chemical.
Raw material for use in the chemical industry.

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Uses advised against : This product must not be used in applications other than the above without first seeking the advice of the supplier.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Biological occupational exposure limits

No biological limit allocated.

8.2 Exposure controls

Monitoring Methods

Monitoring of the concentration of substances in the breathing zone of workers or in the general workplace may be required to confirm compliance with an OEL and adequacy of exposure controls. For some substances biological monitoring may also be appropriate.

Validated exposure measurement methods should be applied by a competent person and samples analysed by an accredited laboratory.

Examples of sources of recommended exposure measurement methods are given below or contact the supplier. Further national methods may be available.

National Institute of Occupational Safety and Health (NIOSH), USA: Manual of Analytical Methods
<http://www.cdc.gov/niosh/>

Occupational Safety and Health Administration (OSHA), USA: Sampling and Analytical Methods
<http://www.osha.gov/>

Health and Safety Executive (HSE), UK: Methods for the Determination of Hazardous Substances
<http://www.hse.gov.uk/>

Institut für Arbeitsschutz Deutschen Gesetzlichen Unfallversicherung (IFA) , Germany
<http://www.dguv.de/inhalt/index.jsp>

L'Institut National de Recherche et de Sécurité, (INRS), France <http://www.inrs.fr/accueil>

Engineering measures

: Use sealed systems as far as possible.
Adequate explosion-proof ventilation to control airborne concentrations below the exposure guidelines/limits.
Local exhaust ventilation is recommended.
Always observe good personal hygiene measures, such as washing hands after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Discard contaminated clothing and footwear that cannot be cleaned.
Practice good housekeeping.
Define procedures for safe handling and maintenance of controls.
Educate and train workers in the hazards and control measures relevant to normal activities associated with this product.
Ensure appropriate selection, testing and maintenance of

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

equipment used to control exposure, e.g. personal protective equipment, local exhaust ventilation.

Drain down system prior to equipment break-in or maintenance.

Retain drain downs in sealed storage pending disposal or for subsequent recycle.

The level of protection and types of controls necessary will vary depending upon potential exposure conditions. Select controls based on a risk assessment of local circumstances. Appropriate measures include:

Personal protective equipment

Protective measures

Personal protective equipment (PPE) should meet recommended national standards. Check with PPE suppliers.

Respiratory protection : If engineering controls do not maintain airborne concentrations to a level which is adequate to protect worker health, select respiratory protection equipment suitable for the specific conditions of use and meeting relevant legislation. Check with respiratory protective equipment suppliers. Where air-filtering respirators are unsuitable (e.g. airborne concentrations are high, risk of oxygen deficiency, confined space) use appropriate positive pressure breathing apparatus. Where respiratory protective equipment is required, use a full-face mask. If air-filtering respirators are suitable for conditions of use: Select a filter suitable for combined particulate/organic gases and vapours [boiling point <65 °C (149 °F)]

Hand protection
Remarks

: If contact with liquefied product is possible or anticipated, gloves should be thermally insulated to prevent cold burns. Where hand contact with the product may occur the use of gloves approved to relevant standards (e.g. Europe: EN374, US: F739) made from the following materials may provide suitable chemical protection. Neoprene rubber. For continuous contact we recommend gloves with breakthrough time of more than 240 minutes with preference for > 480 minutes where suitable gloves can be identified. For short-term/splash protection we recommend the same but recognize that suitable gloves offering this level of protection may not be available and in this case a lower breakthrough time maybe acceptable so long as appropriate maintenance and replacement regimes are followed. Glove thickness is not a good predictor of glove resistance to a chemical as it is dependent on the exact composition of the glove material. Suitability and durability of a glove is dependent on usage, e.g. frequency and duration of contact, chemical resistance of glove material, dexterity. Always seek advice from glove suppliers. Contaminated gloves should be replaced. Personal

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

hygiene is a key element of effective hand care. Gloves must only be worn on clean hands. After using gloves, hands should be washed and dried thoroughly. Application of a non-perfumed moisturizer is recommended.

- Eye protection : Wear safety glasses and face shield (preferably with a chin guard) if splashes are likely to occur.
- Skin and body protection : Chemical and cryogenic gloves/gauntlets, boots, and apron. Wear antistatic and flame-retardant clothing, if a local risk assessment deems it so.
- Thermal hazards : When handling cold material that can cause frost burns, wear cryogenic gloves, safety hat and visor, cold resistant overalls (with cuffs over gloves and legs over boots) and heavy duty boots e.g. leather for cold resistance.
- Hygiene measures : Wash hands before eating, drinking, smoking and using the toilet.
Launder contaminated clothing before re-use.

Environmental exposure controls

- General advice : Local guidelines on emission limits for volatile substances must be observed for the discharge of exhaust air containing vapour.
Information on accidental release measures are to be found in section 6.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

- Appearance : Gas at Standard Temperature and Pressure.
- Colour : colourless
- Odour : Data not available
- Odour Threshold : 270 - 600 ppm
- pH : Not applicable
- Melting point/freezing point : -169,2 °C / -272,6 °F
- Boiling point/boiling range : -103,7 °C / -154,7 °F
- Flash point : -136 °C / -213 °F
Method: No information available.
- Evaporation rate : Data not available
- Flammability (solid, gas) : Flammable gas.

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Upper explosion limit	: 36 %(V)
Lower explosion limit	: 2,7 %(V)
Vapour pressure	: 4.275 kPa (1,9 °C / 35,4 °F)
Relative vapour density	: 0,975 (0 °C / 32 °F)
Relative density	: 0,568 (-104 °C / -155 °F) Method: ASTM D4052
Density	: 568 kg/m3 (-104 °C / -155 °F) Method: ASTM D4052
Solubility(ies)	
Water solubility	: 131 mg/l (25 °C / 77 °F)
Partition coefficient: n-octanol/water	: log Pow: 1,13 Method: Literature data.
Auto-ignition temperature	: 450 °C / 842 °F
Decomposition temperature	: Data not available
Viscosity	
Viscosity, dynamic	: Data not available
Viscosity, kinematic	: Data not available
Particle size	: Data not available

9.2 Other information

Explosive properties	: No data available
Oxidizing properties	: Data not available
Surface tension	: Data not available
Conductivity	: Low conductivity: < 100 pS/m, The conductivity of this material makes it a static accumulator., A liquid is typically considered nonconductive if its conductivity is below 100 pS/m and is considered semi-conductive if its conductivity is below 10,000 pS/m., Whether a liquid is nonconductive or semi-conductive, the precautions are the same., A number of factors, for

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

	example liquid temperature, presence of contaminants, and anti-static additives can greatly influence the conductivity of a liquid
Molecular weight	: 28 g/mol

10. STABILITY AND REACTIVITY

10.1 Reactivity

The product does not pose any further reactivity hazards in addition to those listed in the following sub-paragraph.

10.2 Chemical stability

No hazardous reaction is expected when handled and stored according to provisions. Reacts violently with strong oxidising agents. Reacts violently with hydrochloric acid, hydrogen bromide and nitrogen oxides.

10.3 Possibility of hazardous reactions

Hazardous reactions : Polymerisation may occur at elevated temperatures.

10.4 Conditions to avoid

Conditions to avoid : Heat, flames, and sparks.
Exposure to air.
In certain circumstances product can ignite due to static electricity.

10.5 Incompatible materials

Materials to avoid : Strong oxidising agents.
Hydrochloric acid, hydrogen bromide and nitrogen oxides.

10.6 Hazardous decomposition products

: Thermal decomposition is highly dependent on conditions. A complex mixture of airborne solids, liquids and gases including carbon monoxide, carbon dioxide, sulphur oxides and unidentified organic compounds will be evolved when this material undergoes combustion or thermal or oxidative degradation.

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Basis for assessment : Information given is based on product testing.
Unless indicated otherwise, the data presented is representative of the product as a whole, rather than for individual component(s).

Information on likely routes of exposure : Inhalation is the primary route of exposure.

Acute toxicity

Components:

ethylene:

Acute inhalation toxicity : LC 50 Rat, male: > 20000 ppm
Exposure time: 4 h
Test atmosphere: gas
Method: Literature data
Remarks: Based on available data, the classification criteria are not met.
High gas concentrations will displace available air; unconsciousness and death may occur suddenly from lack of oxygen.

Skin corrosion/irritation

Components:

ethylene:

Remarks: Rapid release of gases which are liquids under pressure may cause frost burns of exposed tissues (skin, eye) due to evaporative cooling.

Serious eye damage/eye irritation

Components:

ethylene:

Remarks: Rapid release of gases which are liquids under pressure may cause frost burns of exposed tissues (skin, eye) due to evaporative cooling.

Respiratory or skin sensitisation

No data available

Germ cell mutagenicity

Components:

ethylene:

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Method: Test(s) equivalent or similar to OECD Guideline 471
Remarks: Based on available data, the classification criteria are not met.

Method: OECD Test Guideline 473
Remarks: Based on available data, the classification criteria are not met.

Test species: Rat
Method: Test(s) equivalent or similar to OECD Test Guideline 474
Remarks: Based on available data, the classification criteria are not met.

Germ cell mutagenicity-Assessment : This product does not meet the criteria for classification in categories 1A/1B.

Carcinogenicity

Components:

ethylene:

Species: Rat, (male and female)

Application Route: Inhalation

Method: Test(s) equivalent or similar to OECD Test Guideline 453

Remarks: Based on available data, the classification criteria are not met.

Carcinogenicity - Assessment : This product does not meet the criteria for classification in categories 1A/1B.

Material	GHS/CLP Carcinogenicity Classification
ethylene	No carcinogenicity classification.

Material	Other Carcinogenicity Classification
ethylene	IARC: Group 3: Not classifiable as to its carcinogenicity to humans

Reproductive toxicity

Components:

ethylene:

Species: Rat

Sex: male and female

Application Route: Inhalation

Method: OECD Test Guideline 421

Remarks: Based on available data, the classification criteria are not met.

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Effects on foetal development

: Species: Rat, male and female
Application Route: Inhalation
Method: OECD Test Guideline 421
Remarks: Based on available data, the classification criteria are not met.

Reproductive toxicity - Assessment

: This product does not meet the criteria for classification in categories 1A/1B.

STOT - single exposure

Components:

ethylene:

Remarks: May cause drowsiness or dizziness., High concentrations may cause central nervous system depression resulting in headaches, dizziness and nausea.

STOT - repeated exposure

Components:

ethylene:

Remarks: Based on available data, the classification criteria are not met.

Repeated dose toxicity

Components:

ethylene:

Rat, male and female:
Application Route: Inhalation
Test atmosphere: Gas
Method: OECD Test Guideline 413
Target Organs: No specific target organs noted
Symptoms: Strain-dependent, Subacute rhinitis, Nasal lesions
Remarks: Treatment-related but slight and not considered adverse.

Aspiration toxicity

11.2 Information on other hazards

Components:

ethylene:

Based on available data, the classification criteria are not met.

Further information

Components:

ethylene:

Remarks: Classifications by other authorities under varying regulatory frameworks may exist.

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

12. ECOLOGICAL INFORMATION

Basis for assessment : Incomplete ecotoxicological data are available for this product. The information given below is based partly on a knowledge of the components and the ecotoxicology of similar products. Unless indicated otherwise, the data presented is representative of the product as a whole, rather than for individual component(s).

12.1 Toxicity

Components:

ethylene :

Toxicity to fish (Acute toxicity) : LC50 : 126,012 mg/l
Exposure time: 96 h
Method: Based on quantitative structure-activity relationship (QSAR) modelling
Remarks: Practically non toxic:
LL/EL/IL50 > 100 mg/l

Toxicity to crustacean (Acute toxicity) : Remarks: Practically non toxic:
LL/EL/IL50 > 100 mg/l

Toxicity to algae/aquatic plants (Acute toxicity) : Remarks: Practically non toxic:
LL/EL/IL50 > 100 mg/l

Toxicity to microorganisms (Acute toxicity) : Remarks: Data not available

Toxicity to fish (Chronic toxicity) : Remarks: Data not available

Toxicity to crustacean(Chronic toxicity) : Remarks: Data not available

12.2 Persistence and degradability

Components:

ethylene :

Biodegradability : Biodegradation: 50 %
Exposure time: 2,9 d
Method: Based on quantitative structure-activity relationship (QSAR) modelling
Remarks: Readily biodegradable.

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

12.3 Bioaccumulative potential

Product:

Partition coefficient: n-octanol/water : log Pow: 1,13Method: Literature data.

Components:

ethylene :

Bioaccumulation : Remarks: Does not bioaccumulate significantly.

12.4 Mobility in soil

Components:

ethylene :

Mobility : Remarks: Because of their extreme volatility, air is the only environmental compartment that hydrocarbon gases will be found.

12.5 Other adverse effects

Product:

Results of PBT and vPvB assessment : The substance does not fulfill all screening criteria for persistence, bioaccumulation and toxicity and hence is not considered to be PBT or vPvB.

Components:

ethylene :

Results of PBT and vPvB assessment : The substance does not fulfill all screening criteria for persistence, bioaccumulation and toxicity and hence is not considered to be PBT or vPvB.

Additional ecological information : In view of the high rate of loss from solution, the product is unlikely to pose a significant hazard to aquatic life.

13. DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Waste from residues	: Recover or recycle if possible. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste classification and disposal methods in compliance with applicable regulations. Do not dispose into the environment, in drains or in water courses. Waste product should not be allowed to contaminate soil or water. Local regulations may be more stringent than regional or national requirements and must be complied with.
Contaminated packaging	: Disposal should be in accordance with applicable regional, national, and local laws and regulations.

14. TRANSPORT INFORMATION

14.1 UN number or ID number

ADR	: 1038
RID	: 1038
IMDG	: 1038
IATA	: 1038 (Not permitted for transport)

14.2 UN proper shipping name

ADR	: ETHYLENE, REFRIGERATED LIQUID
RID	: ETHYLENE, REFRIGERATED LIQUID
IMDG	: ETHYLENE, REFRIGERATED LIQUID
IATA	: ETHYLENE, REFRIGERATED LIQUID

14.3 Transport hazard class(es)

ADR	: 2
RID	: 2
IMDG	: 2.1
IATA	: 2.1 Not permitted for transport

14.4 Packing group

ADR	
Packing group	: Not assigned by regulation
Classification Code	: 3F
Hazard Identification Number	: 223
Labels	: 2.1

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

RID

Packing group : Not assigned by regulation
Classification Code : 3F
Hazard Identification Number : 223
Labels : 2.1

IMDG

Packing group : Not assigned by regulation
Labels : 2.1

IATA

Packing group : Not Assigned
Labels : 2.1

14.5 Environmental hazards

ADR

Environmentally hazardous : no.

RID

Environmentally hazardous : no.

IMDG

Marine pollutant : no.

14.6 Special precautions for user

Remarks : Special Precautions: Refer to Section 7, Handling & Storage, for special precautions which a user needs to be aware of or needs to comply with in connection with transport.

14.7 Maritime transport in bulk according to IMO instruments

Ship type : 2G
Product name : ETHYLENE

Additional Information : Transport in bulk according to the IGC code

This product may be transported under nitrogen blanketing. Nitrogen is an odourless and invisible gas. Exposure to nitrogen enriched atmospheres displaces available oxygen which may cause asphyxiation or death. Personnel must observe strict safety precautions when involved with a confined space entry.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The regulatory information is not intended to be comprehensive. Other regulations may apply to this material.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

Other international regulations

The components of this product are reported in the following inventories:

AU AIIC	: Listed
CA. DSL	: Listed
CN IECSC	: Listed
JP ENCS	: Listed
KR KECI	: Listed
NZ NZIoC	: Listed
PH PICCS	: Listed
TW TCSI	: Listed
US TSCA	: Listed

16. OTHER INFORMATION

Full text of H-Statements

H220	Extremely flammable gas.
H280	Contains gas under pressure; may explode if heated.
H336	May cause drowsiness or dizziness.

Full text of other abbreviations

Flam. Gas	Flammable gases
Press. Gas	Gases under pressure
STOT SE	Specific target organ toxicity - single exposure

Abbreviations and Acronyms : The standard abbreviations and acronyms used in this document can be looked up in reference literature (e.g. scientific dictionaries) and/or websites.

Further information

Training advice : Provide adequate information, instruction and training for operators.

Other information : A vertical bar (|) in the left margin indicates an amendment from the previous version.

Sources of key data used to compile the Safety Data Sheet : The quoted data are from, but not limited to, one or more sources of information (e.g. toxicological data from Shell Health Services, material suppliers' data, CONCAWE, EU IUCLID date base, EC 1272 regulation, etc).

SAFETY DATA SHEET.

Ethylene

Version 1.0

Revision Date. 12.03.2026

Print Date. 13.03.2026

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.